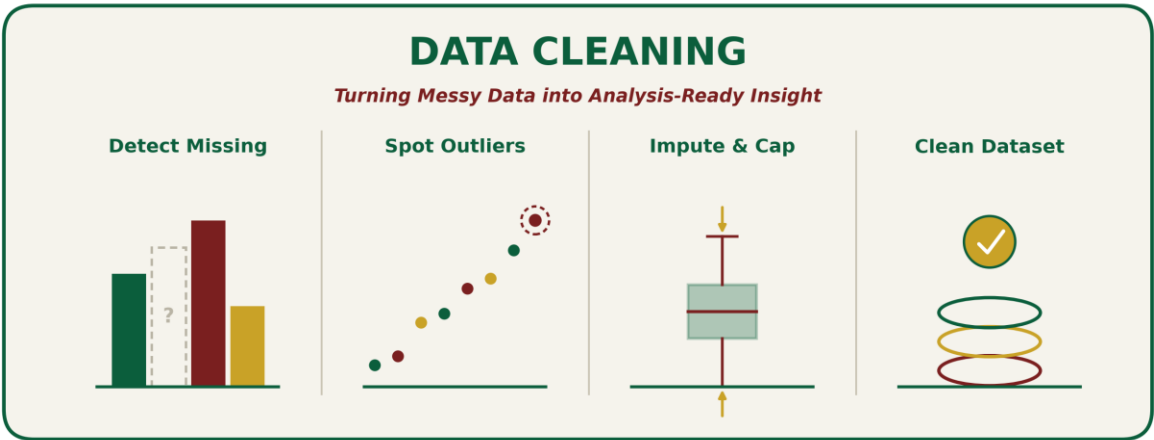


PART C

Final Clean Dataset

Data Cleaning Theory, Diagrams & Report — Patient Health Records Dataset



1. Introduction

The `patient_health_records_dataset.csv` file contains 500 patient records described by 9 columns: `patient_id`, `age`, `gender`, `region`, `bmi`, `blood_pressure`, `cholesterol`, `glucose`, and `disease_risk`. Before this data can be used for statistical analysis or machine learning, it must be cleaned — that is, missing values must be filled in appropriately and extreme/unusual values (outliers) must be handled so that they do not distort the results.

This report explains the theory behind missing value imputation and outlier treatment, shows the methods applied step-by-step with supporting diagrams, and presents the final cleaned dataset along with a brief evaluation of which techniques worked best.

1.1 Data Cleaning Workflow

The diagram below summarizes the end-to-end process followed to convert the raw, incomplete dataset into the final clean dataset used for analysis.

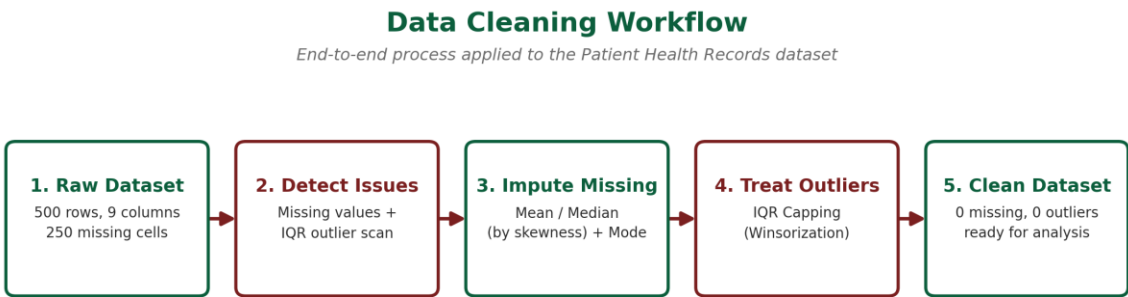


Figure 0: Overall data cleaning workflow — from raw data to final clean dataset.

2. Theory: Handling Missing Values

2.1 What are Missing Values?

Missing values occur when no data value is stored for a variable in an observation. They usually appear as NaN, blank cells, or null entries and can arise due to data entry errors, equipment failure, non-response, or merging of multiple data sources.

2.2 Types of Missingness

- **MCAR (Missing Completely at Random):** The missingness has no relationship with any variable — purely random.
- **MAR (Missing at Random):** The missingness is related to other observed variables, but not the missing value itself.
- **MNAR (Missing Not at Random):** The missingness is related to the value that is missing (e.g., high-income patients not disclosing income).

2.3 Common Imputation Strategies

Imputation Methods Used in This Report

- **Mean Imputation:** Replaces missing values with the column average. Best suited for numeric columns that are roughly symmetric (low skewness), since the mean is sensitive to outliers.
- **Median Imputation:** Replaces missing values with the middle value of the sorted column. Best suited for skewed numeric columns, since the median is resistant to extreme values.
- **Mode Imputation:** Replaces missing values with the most frequently occurring category. Used for categorical columns such as gender and region.

Choosing between mean and median depends on the skewness of the distribution. A skewness value close to 0 indicates a fairly symmetric (near-normal) distribution, where the mean is a reliable measure of central tendency. A skewness value greater than 1 (in magnitude) indicates a strongly skewed distribution, where the median better represents the 'typical' value because it is not pulled by extreme observations.

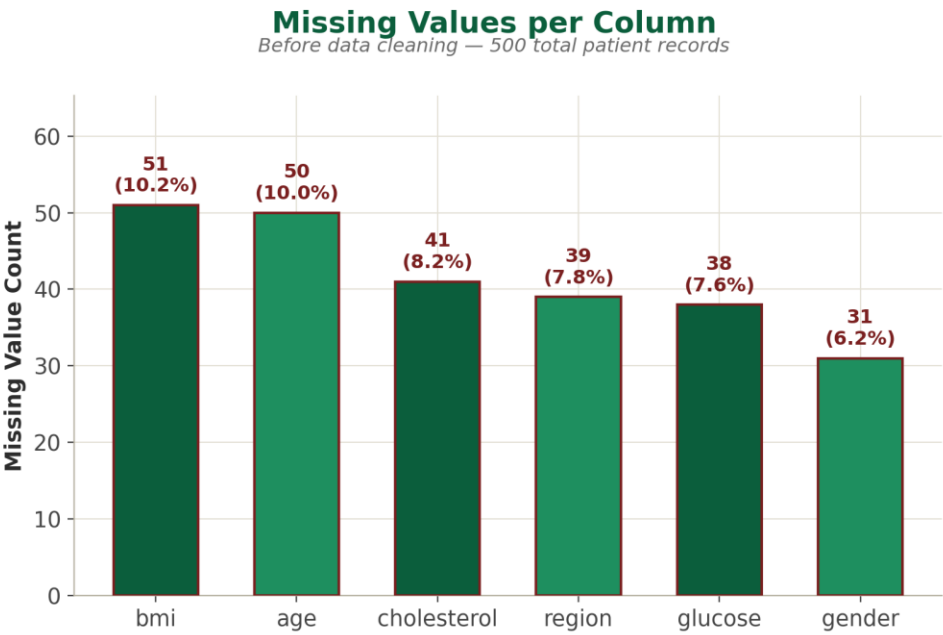


Figure 1: Count of missing values in each column before cleaning.

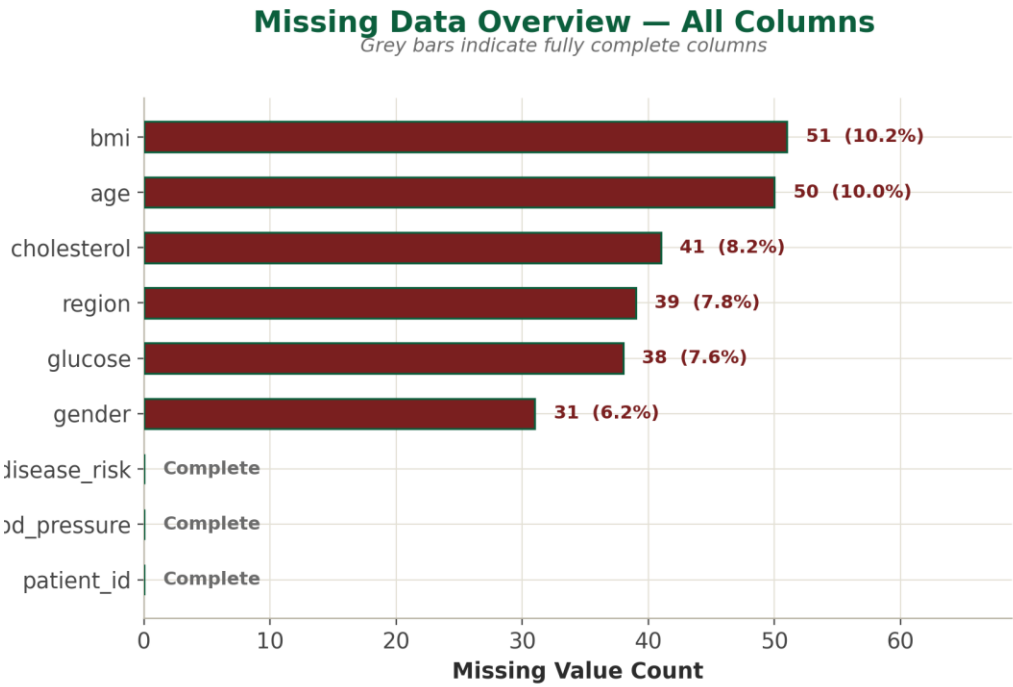


Figure 2: Missing value overview across all columns (grey = complete column).

2.4 Skewness-Based Strategy Applied

Column	Missing Count	Skewness	Strategy Chosen
age	50	0.06 (near symmetric)	Mean
bmi	51	1.24 (skewed)	Median
cholesterol	41	2.46 (skewed)	Median

Column	Missing Count	Skewness	Strategy Chosen
glucose	38	3.09 (highly skewed)	Median
gender / region	31 / 39	Categorical	Mode

Note: blood_pressure had no missing values in the raw file, so no imputation was required for it.

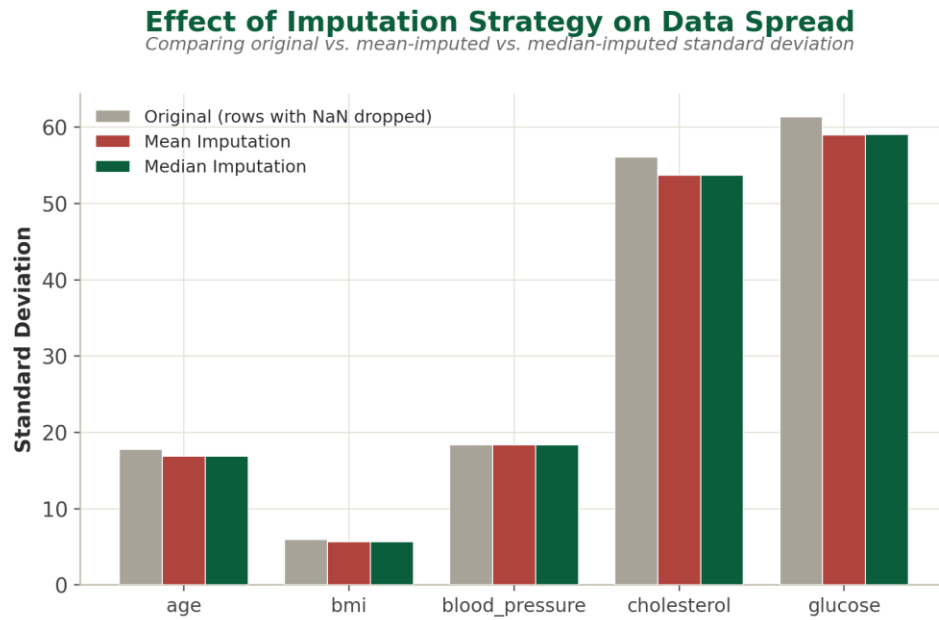


Figure 3: Standard deviation of each numeric column under mean vs. median imputation, compared to the original (rows with missing values dropped).

3. Theory: Handling Outliers

3.1 What are Outliers?

An outlier is a data point that lies far away from the majority of observations in a dataset. Outliers can be genuine extreme measurements (e.g., a very high cholesterol level) or errors in data entry. Left untreated, they can distort the mean, inflate the standard deviation, and mislead any model trained on the data.

3.2 Detection Method: Interquartile Range (IQR)

The IQR method was used to detect outliers, since it does not assume a normal distribution and works well on skewed medical data such as cholesterol and glucose.

IQR Rule

- $Q1 = 25\text{th percentile}$, $Q3 = 75\text{th percentile}$ of the column.
- $IQR = Q3 - Q1$
- Lower Bound = $Q1 - 1.5 \times IQR$
- Upper Bound = $Q3 + 1.5 \times IQR$
- Any value below the Lower Bound or above the Upper Bound is flagged as an outlier.

3.3 Outlier Treatment Methods (Compared)

- **Removal (Trimming):** Deletes outlier rows entirely. Simple, but reduces sample size and can discard genuine extreme patients (e.g., very high-risk cases).
- **Capping / Winsorization:** Replaces values beyond the IQR bounds with the bound value itself, keeping the row but limiting its influence.
- **Transformation (e.g., log transform):** Compresses large values on a log scale — reduces skew but changes the interpretation of the variable.

For this dataset, capping (winsorization) was selected as the final method — it preserves all 500 patient records (important for disease-risk analysis, where every patient matters) while still preventing extreme values from dominating the statistics.

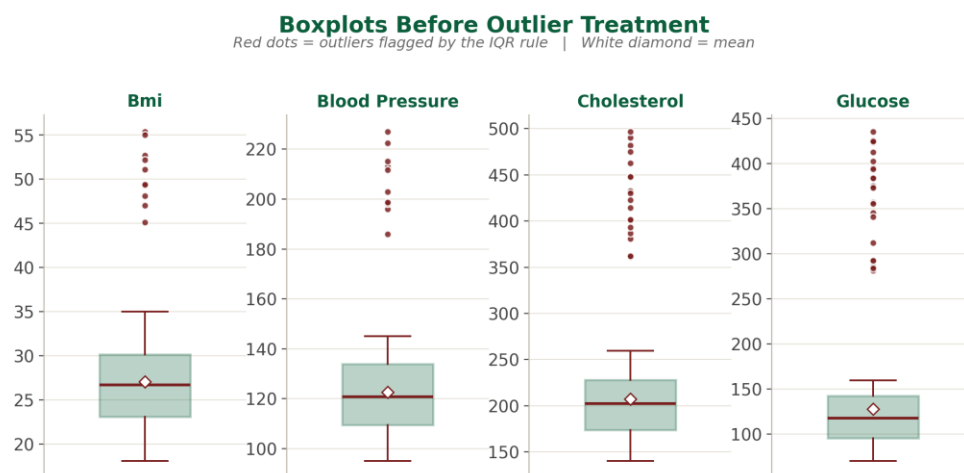


Figure 4: Boxplots of key numeric columns before outlier treatment (red dots are outliers; white diamond is the mean).

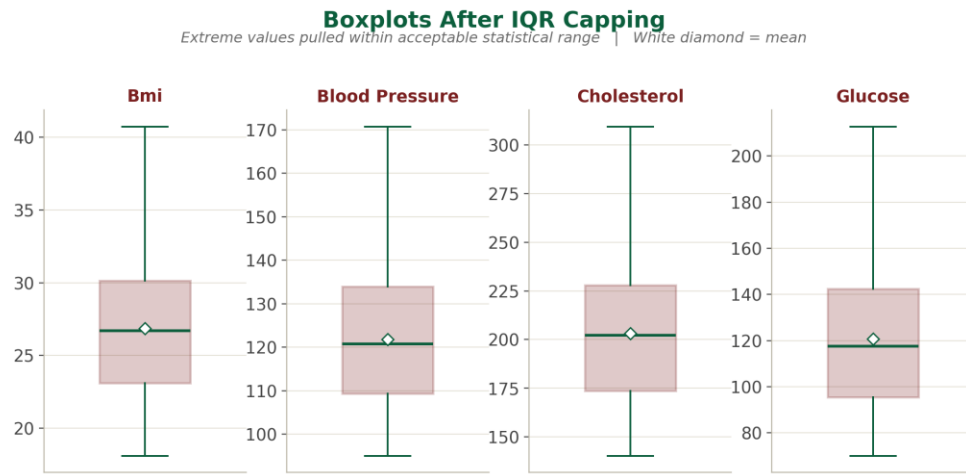


Figure 5: Same columns after IQR capping — outliers pulled within the acceptable range.

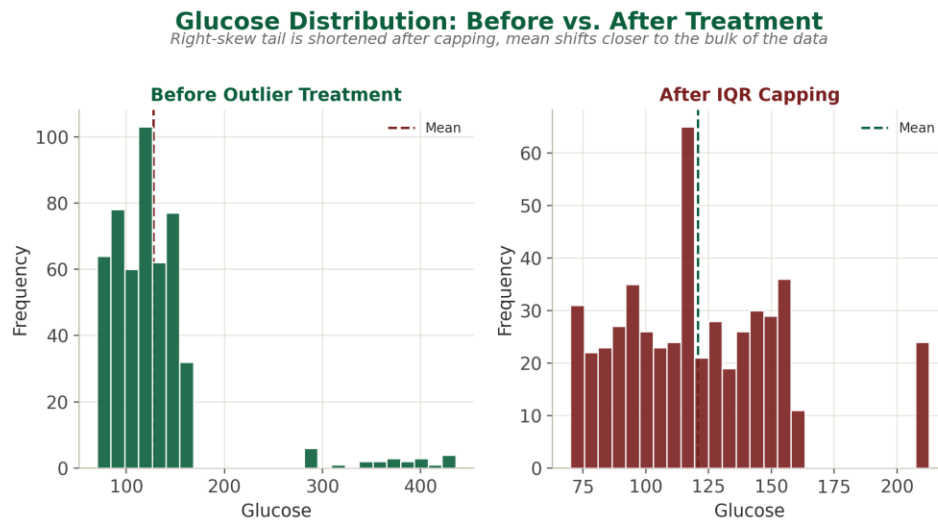


Figure 6: Glucose distribution before and after outlier capping — the extreme right tail is shortened.

3.4 Outlier Bounds Applied

Column	Lower Bound	Upper Bound	Outliers Found
bmi	12.5	40.7	10
blood pressure	72.5	170.8	10
cholesterol	92.1	309.6	18
glucose	25.1	212.8	24

4. Final Cleaned Dataset

After applying skewness-based imputation (Section 2) and IQR-based capping (Section 3), the dataset was rechecked for missing values and out-of-range values.

Cleaning Summary

- Total records: 500 (no rows dropped)
- Total missing values before cleaning: 250 cells
- Total missing values after cleaning: 0 cells
- Total outliers detected: 62 (across bmi, blood_pressure, cholesterol, glucose)
- Total outliers after capping: 0 (all extreme values bounded within IQR limits)

A preview of the final cleaned dataset (first 8 records) is shown below. The complete cleaned file has been saved as `patient_health_records_cleaned.csv`.

ID	Age	Gender	Region	BMI	BP	Cholesterol	Glucose	Risk
1	60	Male	North	27.4	107.2	202.2	79.2	1
2	65	Female	South	25.6	108.9	244.3	138.3	1
3	22	Female	North	34.5	113.9	202.2	144.6	1
4	26	Female	East	27.4	136.7	159.5	102.0	0
5	79	Male	East	32.3	133.8	167.5	72.9	1
6	36	Male	South	30.7	121.9	212.6	108.6	1
7	58	Male	West	24.5	144.8	203.5	157.4	1
8	66	Female	South	26.6	100.3	215.0	145.8	1

5. Brief Report

5.1 Which Imputation Strategy Was Most Effective?

Median imputation was the most effective strategy for bmi, cholesterol, and glucose. As Figure 3 shows, these columns have high skewness, so mean imputation pulls the average toward the extreme tail and slightly inflates the standard deviation, while median imputation keeps the spread closer to the original, missing-free distribution. For age, which is nearly symmetric (skewness ≈ 0.06), mean imputation was suitable and computationally simpler. For the categorical columns gender and region, mode imputation was the only sensible option, since these columns are non-numeric.

5.2 Which Outlier Handling Method Preserved Data Quality Best?

IQR-based capping (winsorization) preserved data quality best compared to removal. Removing all 62 outlier rows would have discarded over 12% of the dataset — a significant loss, especially since many high-value patients (e.g., high cholesterol or glucose) are precisely the ones relevant to predicting disease_risk. Capping retained every patient record while still pulling extreme values within a statistically reasonable range, as seen in the before/after boxplots (Figures 4 and 5).

5.3 How Did Data Cleaning Improve Dataset Usability?

- **Completeness:** All 250 missing cells across 6 columns were filled, so every one of the 500 records can now be used in analysis without being dropped.
- **Reliability:** Skew-aware imputation (mean vs. median) preserved the true shape of each column's distribution instead of introducing artificial bias.
- **Stability:** Capping outliers reduced the influence of extreme values on statistics like mean and standard deviation, making downstream models less sensitive to noise.
- **Consistency:** The final dataset has zero missing values and zero IQR-flagged outliers, making it directly ready for exploratory data analysis, statistical testing, or machine learning.

Conclusion

- The combination of skewness-based imputation and IQR capping transformed a dataset with 250 missing values and 62 outliers into a complete, stable, and analysis-ready dataset of 500 records with 0 missing values and 0 extreme outliers.

6. Overall Workflow

The diagram below consolidates the complete data-cleaning pipeline in one view — the main 5-stage flow, the skewness-based decision logic used for imputation, and the final before/after outcome.

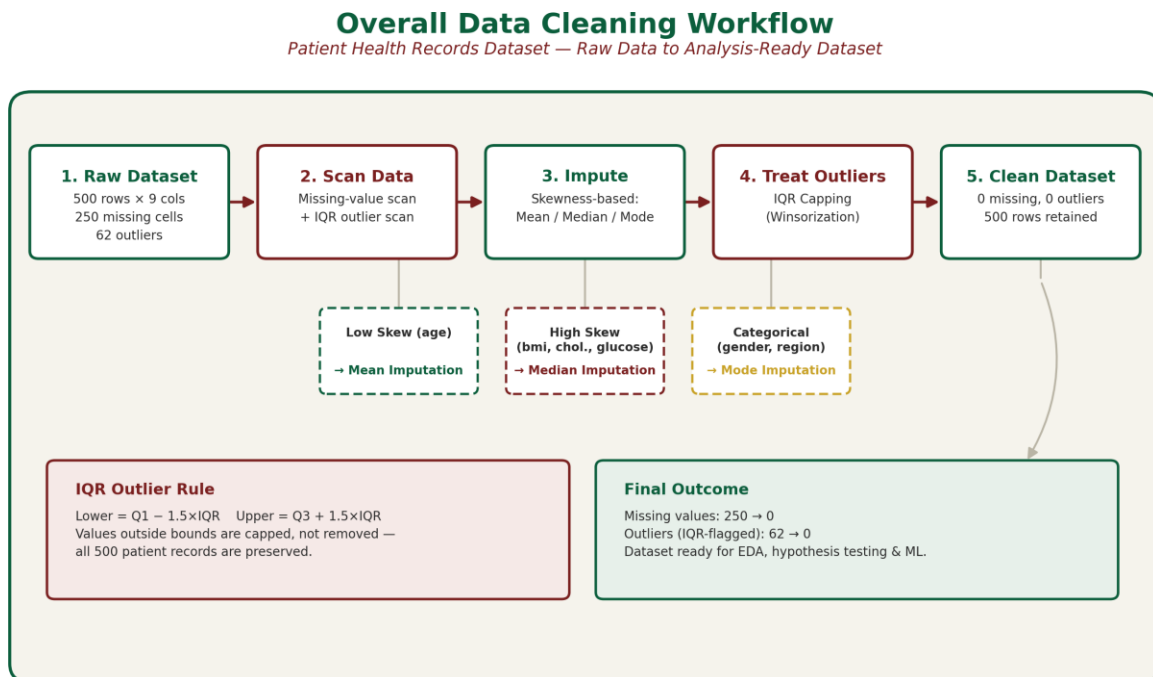


Figure 7: Overall data cleaning workflow, from raw dataset to analysis-ready clean dataset.